INCH-POUND

MIL-C-55021/2B
16 April 1991
SUPERSEDING
MIL-C-55021/2A
8 November 1973

MILITARY SPECIFICATION SHEET

CABLE, ELECTRICAL, SHIELDED SINGLES, SHIELDED AND JACKETED SINGLES, TWISTED PAIRS AND TRIPLES, INTERNAL HOOKUP, -65 TO 200°C

This Specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of the specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-C-55021.

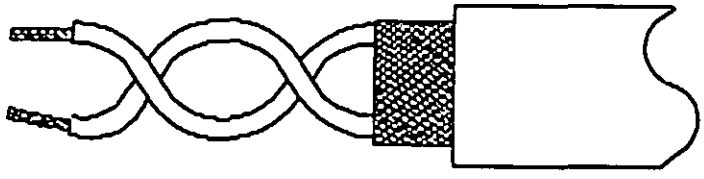


FIGURE 1. Shielded and jacketed cable.

AMSC N/A FSC 6145
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REQUIREMENTS:

Cable construction: The cable shall consist of a shielded singles, shielded and jacketed singles, twisted pairs or triples in accordance with the provisions of MIL-C-55021 (see figure 1 for example).

Type of wire(s): The insulated wires used in constructing the cable shall conform to MIL-W-16878/4 (type E), MIL-W-16878/5 (type EE), as specified (see 6.2 of MIL-C-55021). All wires shall be of the same wire type and AWG size.

Conductor stranding: The conductors stranding shall be in accordance with table I.

TABLE I. Conductor stranding.

AWG size	Number of strands	Nominal stranding 1/	Nominal diameter over conductor 1/	Maximum conductor dc resistance @ 20°C per 1000 ft. (ohms)
32	7	.0031 (0.0787)	.010 (0.254)	173.0
30	7	.0040 (0.1016)	.012 (0.305)	100.7
28	7	.0050 (0.1270)	.015 (0.381)	63.8
26	7	.0063 (0.1600)	.019 (0.483)	40.5
24	19	.0050 (0.1270)	.024 (0.610)	24.3
22	19	.0063 (0.1600)	.030 (0.762)	15.1
20	19	.0080 (0.2032)	.038 (0.965)	9.19
18	19	.0100 (0.2540)	.048 (1.219)	5.79
16	19	.0113 (0.2870)	.057 (1.448)	4.52

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TABLE I. Conductor stranding - Continued.

AWG size	Number of strands	Nominal stranding 1/	Nominal diameter over conductor 1/	Maximum conductor dc resistance @ 20°C per 1000 ft. (ohms)
14	19	.0142 (0.3607)	.071 (1.803)	2.88
12	19	.0179 (0.4547)	.090 (2.286)	1.81
10	37	.0159 (0.4039)	.119 (3.023)	0.106
8	133	.0113 (0.2870)	.166 (4.115)	0.658
6	133	.0142 (0.3607)	.208 (5.283)	0.418

1/ Dimensions are in inches. Metric equivalents, to the nearest 0.1 millimeter (mm), are given for information only and are based upon 1 inch = 25.4 mm.

Length of lay: The length of twist shall be 8 to 16 times the outside major axis diameter of the unshielded, unjacketed cable.

Shield coverage: When specified, (see 1.2.1.5 of MIL-C-55021), the minimum allowable coverage shall be 90 percent, and the angle made by the shielding braid with the longitudinal axis of the twisted conductors shall be between 20 and 40 degrees.

Jacket materials: When specified, (see 1.2.1.5 of MIL-C-55021), the cable jacket shall be extruded or tape wrapped polytetrafluoroethylene (PTFE), or fluorinated ethylene propylene (FEP) of uniform thickness as specified in table II. If the jacket is tape, two layers of tape shall be applied in opposite directions, each layer and adjacent convolutions being properly sealed together to form a homogeneous wall. The surface of the jacket shall be smooth, although a slight spiral ridge of the tape shall not be cause for rejection. Prefused tape shall not be used.

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Table II. Jacket thickness.

Cable diameter (under jacket) 1/ 2/	Nominal jacket thickness 2/	Minimum jacket thickness 2/
Under .300 (7.62)	.012 (0.305)	.008 (0.203)
.300 and over	.016 (0.406)	.012 (0.305)

1/ When specifying a jacket thickness both unshielded and shielded constructions are applicable.
2/ Dimensions are in inches. Metric equivalents, to the nearest 0.1 millimeter (mm), are given for information only and are based upon 1 inch = 25.4 mm.

Heat resistance: The cable shall be tested at 250°C +/- 5°C. (482°F +/- 9°F) air oven temperature for 96 hours.

Type designation: The type designation shall be in accordance with the provisions of this specification sheet and MIL-C-55021 (see 1.2.1 of MIL-C-55021).

CONCLUDING MATERIAL

Custodians:
Army - MI
Navy - AS
Air Force - 85

Review activities:
Army - CR
DLA - IS

Preparing activity:
Air Force - 85

Agent:
DLA - ES

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